



IT-SIMPLERA INSTITUTE
Cyber Security Analyst Internship Program

WEEK 01 ASSIGNMENT REPORT
Cybersecurity Fundamentals & Security Concepts

Prepared By:	Syed Hussain Ali Shirazi
Assigned To:	Cyber Security Analyst Interns
Department:	Cyber Security
Supervisor:	Senior Cyber Security Analyst
Submission Date:	3 July 2026

Assignment Objectives: Develop cybersecurity fundamentals; Understand the CIA Triad; Research real-world cyberattacks; Learn common cybersecurity tools; Design a secure home network; Document daily responsibilities of a Cybersecurity Analyst.



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1. INTRODUCTION

Cybersecurity protects systems, networks, and data from unauthorized access and attacks. This assignment covers core concepts including the **CIA Triad** (Confidentiality, Integrity, Availability), real-world cyberattacks, essential tools, secure network design, and the daily role of a cybersecurity analyst.

CIA Triad: **Confidentiality** ensures only authorized access (encryption, MFA); **Integrity** keeps data accurate and unaltered (hashing, signatures); **Availability** ensures systems are accessible when needed (backups, redundancy). Most attacks violate one or more of these principles, guiding defensive priorities.

This report addresses all five assignment tasks with concise analysis and practical deliverables.

2. TASK 1 & 2: REAL-WORLD CYBERATTACKS & CIA TRIAD ANALYSIS

Five major attacks analyzed through the CIA Triad lens:

2.1 WannaCry Ransomware (2017) — NHS UK + 200k systems globally. Self-propagating worm via EternalBlue exploit. Encrypted files, demanded Bitcoin. ~19k appointments cancelled, £92M NHS cost. **CIA:** Availability (primary — hospitals crippled) + Integrity (files encrypted). Lesson: Patch management critical.

2.2 Equifax Breach (2017) — 147.9M Americans affected via unpatched Apache Struts. Names, SSNs, DLs, credit cards stolen. \$575-700M settlement. **CIA:** Confidentiality (massive PII exfiltration). Lesson: Patch known vulnerabilities immediately.

2.3 SolarWinds Supply Chain (2020) — Malicious backdoor in Orion updates affected 18k+ orgs (US gov, Microsoft, FireEye). Stealthy espionage. **CIA:** Confidentiality (long-term data theft) + Integrity (compromised updates). Lesson: Verify software supply chain & code signing.

2.4 Colonial Pipeline (2021) — DarkSide ransomware via weak VPN password. 6-day US East Coast fuel pipeline shutdown, panic buying, \$4.4M ransom. **CIA:** Availability (critical infra offline) + Confidentiality/Integrity. Lesson: MFA everywhere + IR planning for critical systems.

2.5 Yahoo Breaches (2013-14) — ~3B accounts compromised (state-sponsored). Emails, hashed passwords, security questions stolen. \$350M Verizon deal haircut. **CIA:** Confidentiality (largest PII breach ever). Lesson: Timely disclosure + secure credential storage.

Key Insight: Most attacks succeed due to basic failures (unpatched systems, weak MFA, poor vendor trust). The CIA Triad helps quickly identify the primary business impact and prioritize controls.

3. TASK 3: CYBERSECURITY TOOLS

Wireshark — Network protocol analyzer for packet capture, deep inspection, and forensics. Use cases: Detect C2 traffic, troubleshoot issues, reconstruct attacker sessions, validate firewall rules.

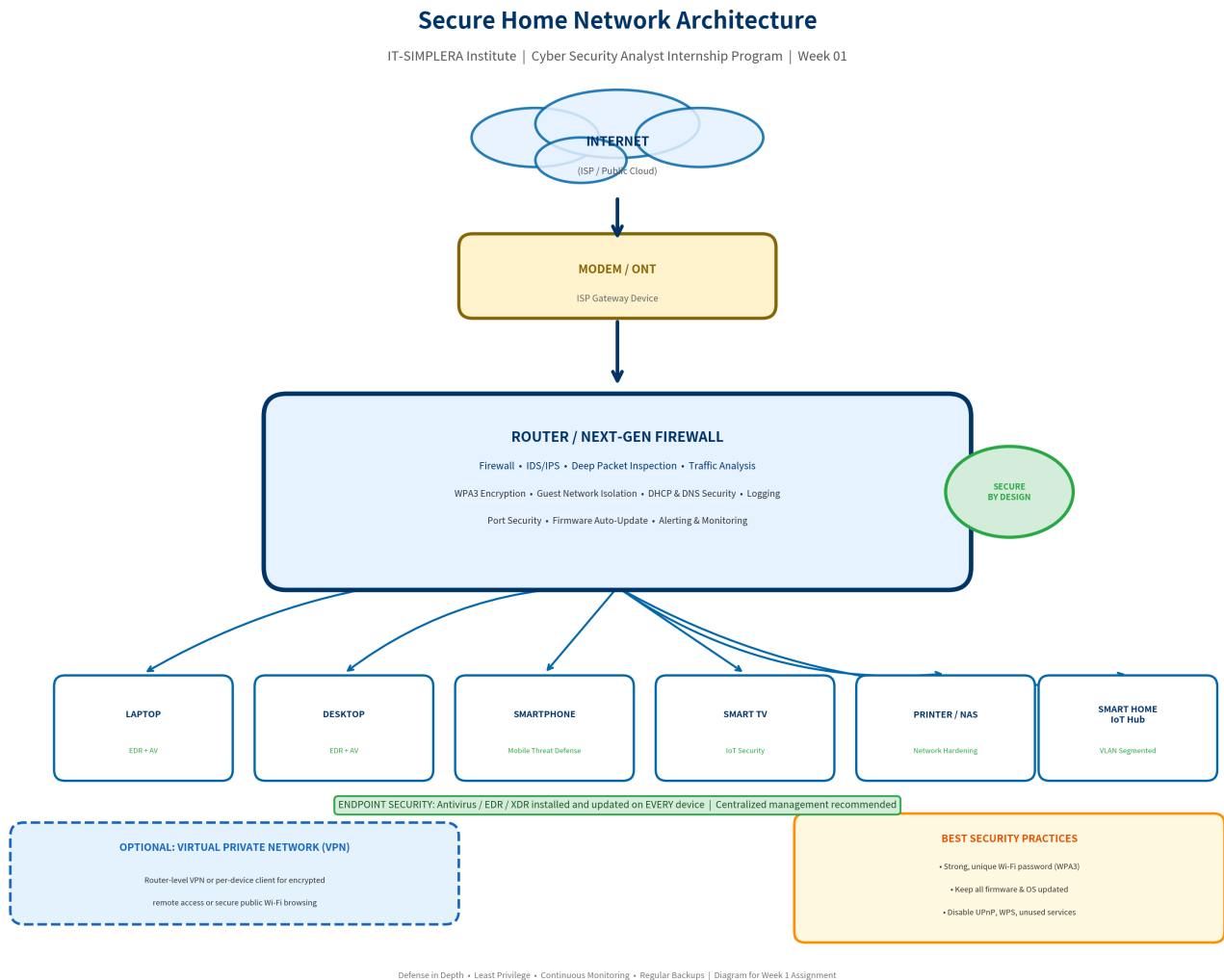
Nmap — Network scanner for host discovery, port scanning, service/OS detection, and vuln scripting (NSE). Use cases: Asset inventory, external attack surface assessment, compliance checks, rogue device detection.

Burp Suite — Web app security testing platform (proxy, scanner, repeater, intruder). Use cases: Web/API pentesting, finding OWASP Top 10 flaws, bug bounties, secure code training.

Why These Matter: Wireshark helps you "see" the network, Nmap helps you "map" it, and Burp Suite helps you "break" web apps — the #1 attack vector. Essential for SOC, pentest, and red team roles.

4. TASK 4: SECURE HOME NETWORK DIAGRAM

A secure home network follows the principle of **defense in depth** — multiple overlapping controls so that compromise of one layer does not lead to total failure. The diagram below illustrates a recommended architecture for a typical household or small office, incorporating the components specified in the assignment.



Key Controls: Next-Gen Firewall + IDS/IPS on router, WPA3 + Guest isolation, EDR/AV on all endpoints, optional VPN, regular updates, network segmentation for IoT, 3-2-1 backups, password manager + MFA. **Defense in Depth** ensures no single failure compromises the network. Similar layered approach scales to enterprise SOCs.



5. TASK 5: DAILY LIFE OF A CYBERSECURITY ANALYST

A Cybersecurity Analyst protects organizational systems and data by identifying, detecting, investigating, and responding to threats. Core mission: prevent incidents before they cause damage. Work often involves shift-based SOC monitoring.

Typical Daily Activities:

- **Alert Triage:** Review SIEM/EDR/firewall alerts, prioritize by risk/asset criticality, dismiss false positives.
- **Incident Response:** Contain (isolate hosts, block IOCs), investigate (log correlation, scope assessment), eradicate, recover, and document.
- **Proactive Work:** Vulnerability management & patching coordination, threat hunting for missed IOCs, tuning detection rules, reading threat intel (Mandiant, CISA).
- **Collaboration:** Communicate risks clearly to IT and business stakeholders; participate in tabletop exercises.
- **Continuous Improvement:** Training, certifications (Security+, CySA+), purple teaming, playbook updates.

Reality: Much of the role is methodical alert review, documentation, and tuning rather than dramatic breaches. Success requires technical skill + clear communication + disciplined processes. Good tooling and work-life balance are essential to avoid burnout in high-volume environments.

6. CHALLENGES & CONCLUSION

Challenges: Selecting accurate sources amid information overload, applying CIA Triad nuances without oversimplification, designing a clean yet complete network diagram, balancing depth with conciseness, and managing time as an intern while maintaining professional standards.

Conclusion: This assignment built a strong foundation in cybersecurity fundamentals. Researching major attacks (WannaCry, Equifax, SolarWinds, Colonial Pipeline, Yahoo) showed how the CIA Triad guides impact assessment. Mastering Wireshark, Nmap, and Burp Suite provided practical tools for seeing, mapping, and testing systems. The secure home network design reinforced defense-in-depth principles. Understanding an analyst's daily responsibilities highlighted the mix of technical rigor, communication, and continuous learning required. I am excited to apply these skills during the internship and grow as a cybersecurity professional.

Next Steps: LinkedIn post on key learnings • GitHub repo with report + diagram • Interview preparation.